

Experiment 7:

Aim: To demonstrate the performance of a logistic regression by using choosen database with python code.

Program:

```
# Import libraries
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
cancer = load_breast_cancer()

# Features and target
X = cancer.data
y = cancer.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Logistic Regression model
model = LogisticRegression(max_iter=5000)

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n", cm)

# Classification Report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

# Plot Confusion Matrix
plt.figure(figsize=(5,4))
sns.heatmap(cm,
            annot=True,
            fmt="d",
            cmap="Blues",
```

```

xticklabels=cancer.target_names,
yticklabels=cancer.target_names)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Logistic Regression - Breast Cancer Dataset")
plt.show()

```

Accuracy: 0.9766081871345029

Confusion Matrix:

```
[[ 61   2]
 [  2 106]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.97	0.97	0.97	63
1	0.98	0.98	0.98	108
accuracy			0.98	171
macro avg	0.97	0.97	0.97	171
weighted avg	0.98	0.98	0.98	171

